CHAPTER 04

Post-Optimal and Sensitivity Analysis

Shadow prices, reduced costs, and sensitivity analysis

TEXTBOOK	CHAPTER	WEEK
Modern Management Science, 9th ed.	Ch. 4	Week 5

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Post-Optimal and Sensitivity Analysis

Post-optimal analysis: What-if analysis performed after optimization

- Evaluating resource values using **shadow prices**
- Evaluating new-product profitability using **reduced costs**
- **Sensitivity analysis** of changes in objective coefficients and resource availability (constraints)

Post-Optimal and Sensitivity Analysis

Shadow price

the increase in the objective function when the supply capacity of a resource increases by 1

Dual variable

mathematically, the shadow price is called the dual variable; it is called the shadow price when it has an economic interpretation

Shadow Price

Example 4-1

Gap Chemical, a small fertilizer company, produces two fertilizers, premium and regular, using three raw materials A, B, and C. For next month, Gap Chemical has secured 1,500 tons of A, 1,200 tons of B, and 500 tons of C. Producing 1 ton of premium fertilizer requires 2 tons of A, 1 ton of B, and 1 ton of C; producing 1 ton of regular fertilizer requires 1 ton each of A and B and no C.

Premium fertilizer sells for ₩700,000 per ton, earning a profit of ₩170,000 after material and other costs; regular fertilizer sells for ₩300,000 per ton, earning a profit of ₩100,000 after material and other costs. Gap Chemical wants to set a production plan that maximizes profit.

Shadow Price

Variables

X_1: premium fertilizer production

X_2: regular fertilizer production

Model

Maximize $Z = 17X_1 + 10X_2$ (profit, ₩10,000)

Subject to $2X_1 + X_2 <= 1500$ (Material A, tons)

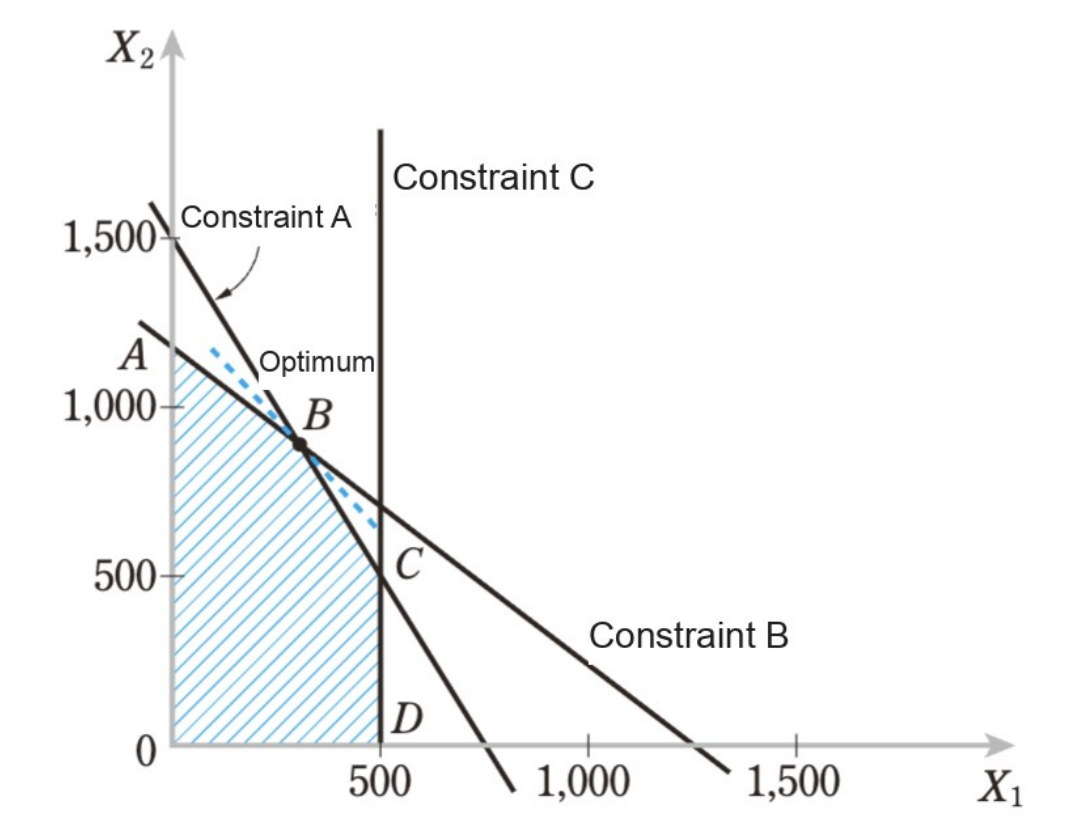
$X_1 + X_2 <= 1200$ (Material B, tons)

$X_1 <= 500$ (Material C, tons)

$X_1, X_2 >= 0$

Shadow Price

Figure 4.1 Graphical analysis of Example 4-1



Optimal solution? Obtained by solving the following

$2X_1 + X_2 = 1500$

$X_1 + X_2 = 1200$

Optimal solution: X_1=300, X_2=900

Optimal value: Z = ₩141 million

$2X_1 + X_2 = 1500$

$X_1 + X_2 = 1200$

X1 = 1200 - X2

2*(1200-X2) + X2 = 1500

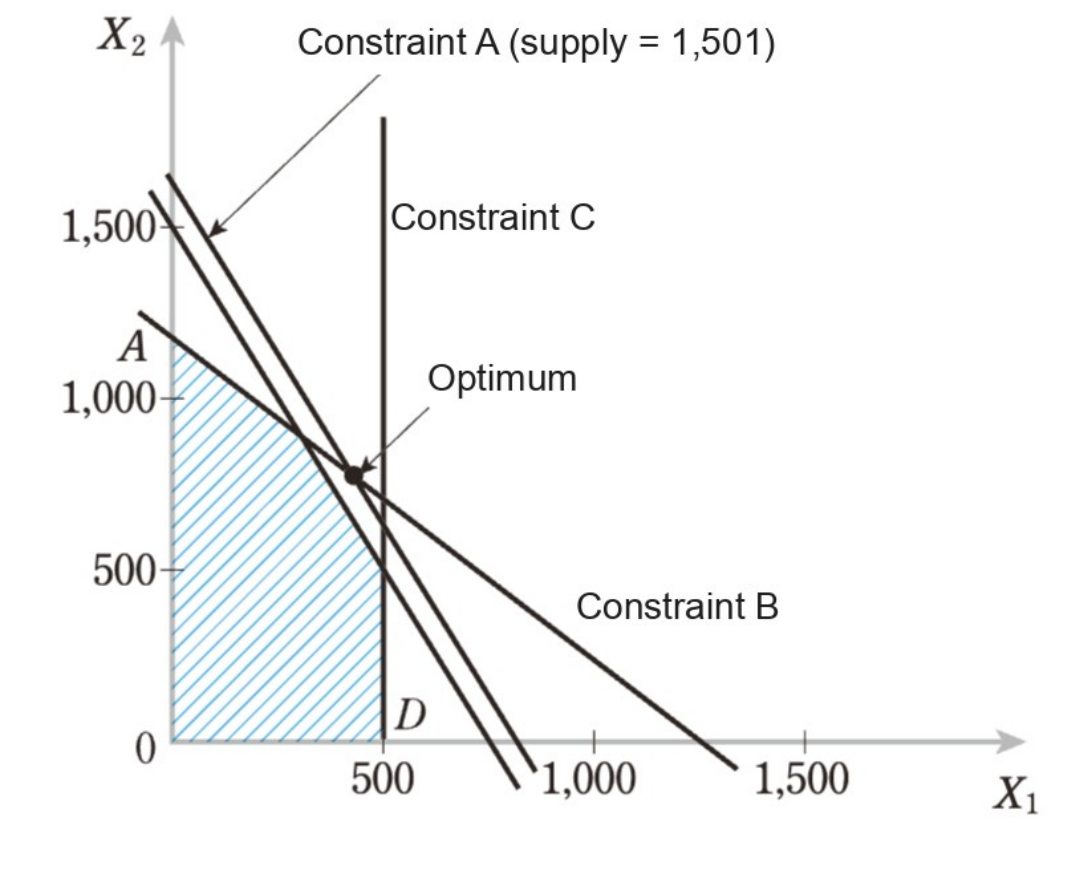
X2 = 900

X1= 300

$Z = 17X_1 + 10X_2 = 17 * 300 + 10 * 900 = 141000$

Shadow Price

Figure 4.2 Material A supply increases by 1 ton



Optimal solution? Obtained by solving the following

$2X_1 + X_2 = 1501$

$X_1 + X_2 = 1200$

Optimal solution: X_1= 301, X_2= 899

Optimal value: Z = ₩141.07 million

Shadow Price

X1 = 1200 – X2 2(1200-X2) + X2 = 1501 X2 = 899 X1 = 301

Original

From the constraints

$2X_1 + X_2 = 1500$

$X_1 + X_2 = 1200$

Opt. solution: X_1= 300, X_2= 900

Opt. value: Z = ₩141 million

Material A supply +1 unit (1 ton)

From the constraints

$2X_1 + X_2 = 1501$

$X_1 + X_2 = 1200$

Opt. solution: X_1= 301, X_2= 899

Opt. value: Z = ₩141.07 million

Material A supply +1 unit (1 ton) → optimal value (i.e., objective function) increases by ₩70,000

Shadow price of Material A = ₩70,000/ton

This also represents the marginal profit of Material A's supply capacity

Shadow Price

Find the shadow prices of Materials B and C in the same way

To find the shadow price of Material B, modify the constraints as follows

< Original >

Maximize $Z = 17X_1 + 10X_2$

Subject to $2X_1 + X_2 <= 1500$ (Material A)

$X_1 + X_2 <= 1200$ (Material B)

$X_1 <= 500$ (Material C)

$X_1, X_2 >= 0$

< Material B capacity +1 unit >

Maximize $Z = 17X_1 + 10X_2$

Subject to $2X_1 + X_2 <= 1500$ (Material A)

$X_1 + X_2 <= 1201$ (Material B)

$X_1 <= 500$ (Material C)

$X_1, X_2 >= 0$

Shadow Price

X1 = 1201 – X2 2(1201-X2) + X2 = 1500 X1, X2 = (299, 902) Z = 17*(299) + 10*(902)

Original

From the constraints

$2X_1 + X_2 = 1500$

$X_1 + X_2 = 1200$

Opt. solution: X_1= 300, X_2= 900

Opt. value: Z = ₩141 million

Material B supply +1 unit (1 ton)

From the constraints

$2X_1 + X_2 = 1500$

$X_1 + X_2 = 1201$

Opt. solution: X_1= 299, X_2= 902

Opt. value: Z = ₩141.03 million

Material B supply +1 unit (1 ton) → optimal value (i.e., objective function) increases by ₩30,000

Shadow price of Material B = ₩30,000/ton

Shadow Price

Find the shadow prices of Materials B and C in the same way

To find the shadow price of Material C, modify the constraints as follows

< Original >

Maximize $Z = 17X_1 + 10X_2$

Subject to $2X_1 + X_2 <= 1500$ (Material A)

$X_1 + X_2 <= 1200$ (Material B)

$X_1 <= 500$ (Material C)

$X_1, X_2 >= 0$

< Material C capacity +1 unit >

Maximize $Z = 17X_1 + 10X_2$

Subject to $2X_1 + X_2 <= 1500$ (Material A)

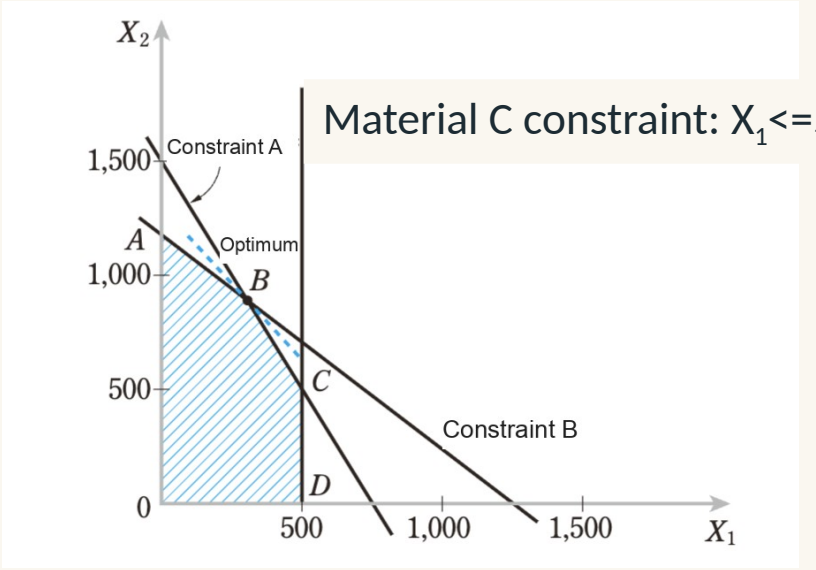
$X_1 + X_2 <= 1200$ (Material B)

$X_1 <= 501$ (Material C)

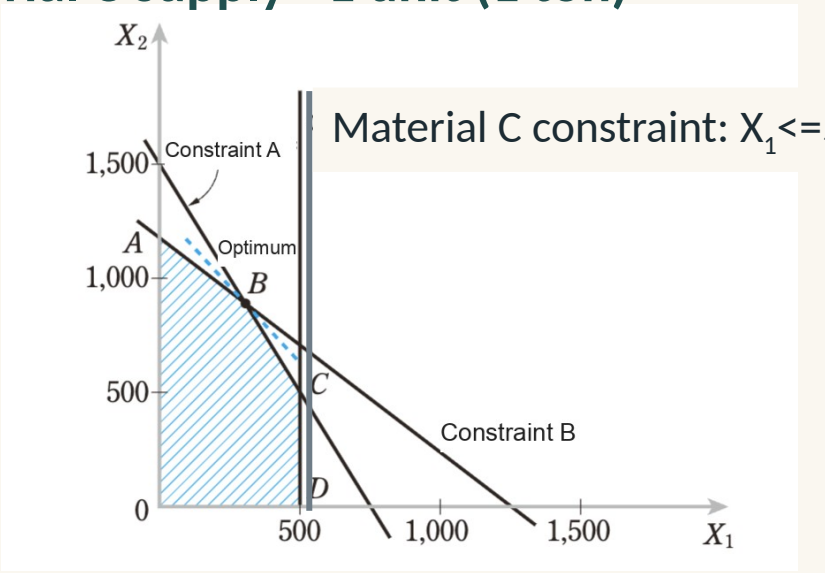
$X_1, X_2 >= 0$

Shadow Price

Original



Material C supply +1 unit (1 ton)



Increasing the supply of Material C does not move the optimal solution (point B)

Material C supply +1 unit (1 ton) → optimal value (i.e., objective function) increases by ₩0

Shadow price of Material C = ₩0/ton

Shadow Price

Summary

Shadow price of Material A = ₩70,000/ton

Shadow price of Material B = ₩30,000/ton

Shadow price of Material C = ₩0/ton

In general, if a constraint is non-binding, its shadow price = 0

Post-Optimal Analysis using Shadow Price

Valuing additional materials

- If the shadow prices of Materials A, B, and C are ₩70,000/ton, ₩30,000/ton, and ₩0/ton, respectively, which material is worth acquiring more of to maximize profit?
- Between acquiring more Material A and more Material B, which is more profitable, and by how much?

Post-Optimal Analysis using Shadow Price

Deciding whether to buy more material

- Company L offers to sell Material A at a ₩40,000 premium over its price. Should we buy it?

Post-Optimal Analysis using Shadow Price

Deciding whether to produce a new product

- A new product requires 1.5 tons of A, 1.5 tons of B, and 1 ton of C. Selling it yields a profit of ₩130,000 after material costs. Is it profitable to produce this new product?

- What is the value of the A, B, C capacity used to make the new product?

Reduced Cost

- **Reduced cost**: the change in the objective function value when a variable that equals 0 in the optimal solution is increased by 1 unit is called the reduced cost (or relative cost).

Reduced Cost

Example 4-2

Gap Chemical from [Example 4-1] now produces three fertilizers: super-premium in addition to premium and regular. Producing 1 ton of super-premium requires 3 tons of A, 1 ton of B, and 1 ton of C. It sells for ₩700,000 per ton, earning a profit of ₩210,000 after material and other costs. Everything else is the same as in [Example 4-1].

Reduced Cost

Example 4-1 (two fertilizers)

Maximize $Z = 17X_1 + 10X_2$ (profit, ₩10,000)

Subject to $2X_1 + X_2 <= 1500$ (Material A, tons)

$X_1 + X_2 <= 1200$ (Material B, tons)

$X_1 <= 500$ (Material C, tons)

$X_1, X_2 >= 0$

Example 4-2

Gap Chemical from [Example 4-1] now produces three fertilizers: super-premium in addition to premium and regular. Producing 1 ton of super-premium requires 3 tons of A, 1 ton of B, and 1 ton of C. It sells for ₩700,000 per ton, earning a profit of ₩210,000 after material and other costs. Everything else is the same as in [Example 4-1].

Example 4-2 (+ super-premium)

Maximize $Z = 17X_1 + 10X_2 + 21X_3$ (profit, ₩10,000)

Subject to $2X_1 + X_2 + 3X_3 <= 1500$ (Material A, tons)

$X_1 + X_2 + X_3 <= 1200$ (Material B, tons)

$X_1 + X_3 <= 500$ (Material C, tons)

$X_1, X_2, X_3 >= 0$

Optimal solution $(X_1, X_2, X_3) = (300, 900, 0)$

Optimal value Z = ₩141 million

Reduced Cost

Example 4-2 (+ super-premium)

Maximize $Z = 17X_1 + 10X_2 + 21X_3$ (profit, ₩10,000)

Subject to $2X_1 + X_2 + 3X_3 <= 1500$ (Material A, tons)

$X_1 + X_2 + X_3 <= 1200$ (Material B, tons)

$X_1 + X_3 <= 500$ (Material C, tons)

$X_1, X_2, X_3 >= 0$

Optimal solution $(X_1, X_2, X_3) = (300, 900, 0)$

Optimal value Z = ₩141 million

Forcing 1 ton of super-premium

Maximize $Z = 17X_1 + 10X_2 + 21*1$ (profit, ₩10,000)

Subject to $2X_1 + X_2 + 3*1 <= 1500$ (Material A, tons)

$X_1 + X_2 + 1 <= 1200$ (Material B, tons)

$X_1 + 1 <= 500$ (Material C, tons)

$X_1, X_2 >= 0$

Optimal solution $(X_1, X_2, X_3) = (298, 901, 1)$

Optimal value Z = ₩140.97 million → profit falls by ₩30,000!

Reduced Cost

Reduced cost

Reduced cost: the change in the objective function value when a variable that equals 0 in the optimal solution is increased by 1 unit

In the example on the right, the reduced cost of X_3 is –₩30,000.

Forcing 1 ton of super-premium

Maximize $Z = 17X_1 + 10X_2 + 21*1$ (profit, ₩10,000)

Subject to $2X_1 + X_2 + 3*1 <= 1500$ (Material A, tons)

$X_1 + X_2 + 1 <= 1200$ (Material B, tons)

$X_1 + 1 <= 500$ (Material C, tons)

$X_1, X_2 >= 0$

Optimal solution $(X_1, X_2, X_3) = (298, 901, 1)$

Optimal value Z = ₩140.97 million → profit falls by ₩30,000!

Sensitivity Analysis

F Electronics, a small electronics company producing standard lamps and high-performance lamps, wants to decide how many of each product to make so as to maximize profit within its current production capacity. Lamp production goes through an assembly process and an inspection process, with 240 and 81 labor-hours available per day, respectively. Each standard lamp requires 1.2 hours of assembly and 0.5 hours of inspection labor; each high-performance lamp requires 4 hours of assembly and 1 hour of inspection labor. In addition, each high-performance lamp requires one special semiconductor chip, but supply of this chip has recently been tight, so only 40 chips can be procured per day. The company's lamps are of good quality and sell out completely; each standard lamp earns a profit of ₩200,000 and each high-performance lamp ₩500,000.

Variables

X_1: daily production of standard lamps

X_2: daily production of high-performance lamps

Model

maximize $Z = 20\,X_1 + 50\,X_2$ (total profit, ₩10,000)

subject to $1.2\,X_1 + 4\,X_2 <= 240$ (assembly, hours)

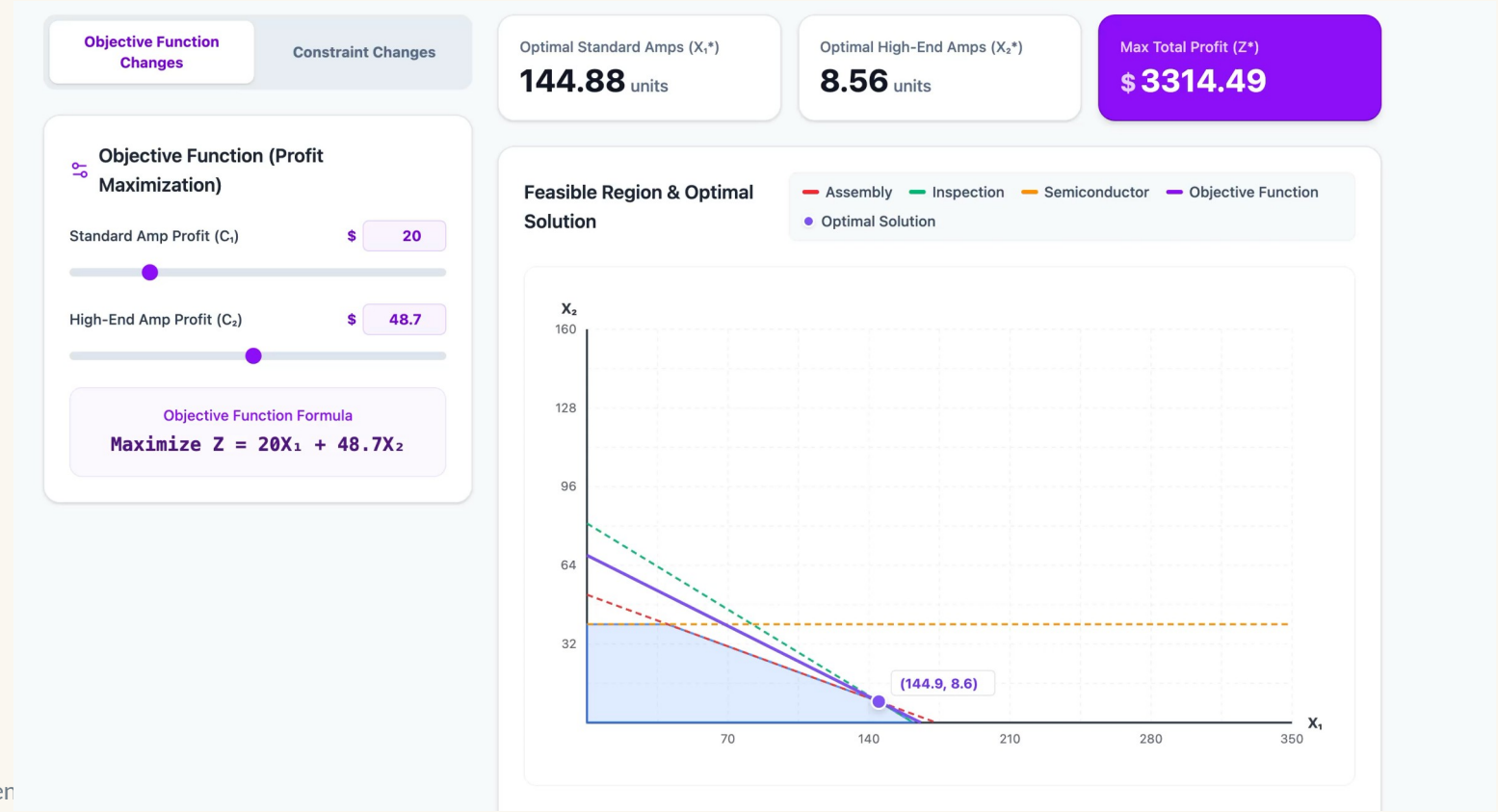
$0.5\,X_1 + X_2 <= 81$ (inspection, hours)

$X_2 <= 40$ (chip supply, units)

$X_1, X_2 >= 0$

Sensitivity Analysis – Objective Coefficients

Mathematically, we can use the property that “the optimal solution is unchanged if the objective’s slope lies between the slopes of the two binding constraints” to find the range over which an objective coefficient can change without changing the optimal solution (the allowable range)



Sensitivity Analysis – Objective Coefficients

Change in objective coefficient C_2

For $40 <= C_2 <= 200/3$, the optimal solution (X_1=105, X_2=28.5) does not change

Here, the optimal value $Z = 20 \times 105 + C_2 \times 28.5 = 28.5C_2 + 2{,}100$ (₩10,000) is proportional to C_2

Sensitivity Analysis – Changes in the RHS

When the right-hand side (RHS) changes

- Changing the RHS shifts the constraint boundary, so the optimal solution itself changes
- Small changes in the RHS are captured by the shadow price
- However, when the change in the RHS is large, the shadow price itself changes
- So, we need to check how far the RHS can change while the shadow price stays the same

Sensitivity Analysis – Changes in the RHS

When the right-hand side (RHS) changes

Objective Function Changes | Constraint Changes

Optimal Standard Amps (X_1*) **105.00** units

Optimal High-End Amps (X_2*) **28.50** units

Max Total Profit (Z*) **$ 3525.00**

Constraints (Available Resources)

Assembly Dept. Available Hours (b_1) 240 hrs

$1.2X_1 + 4X_2 \leq 240$ — Shadow Price: $6.25

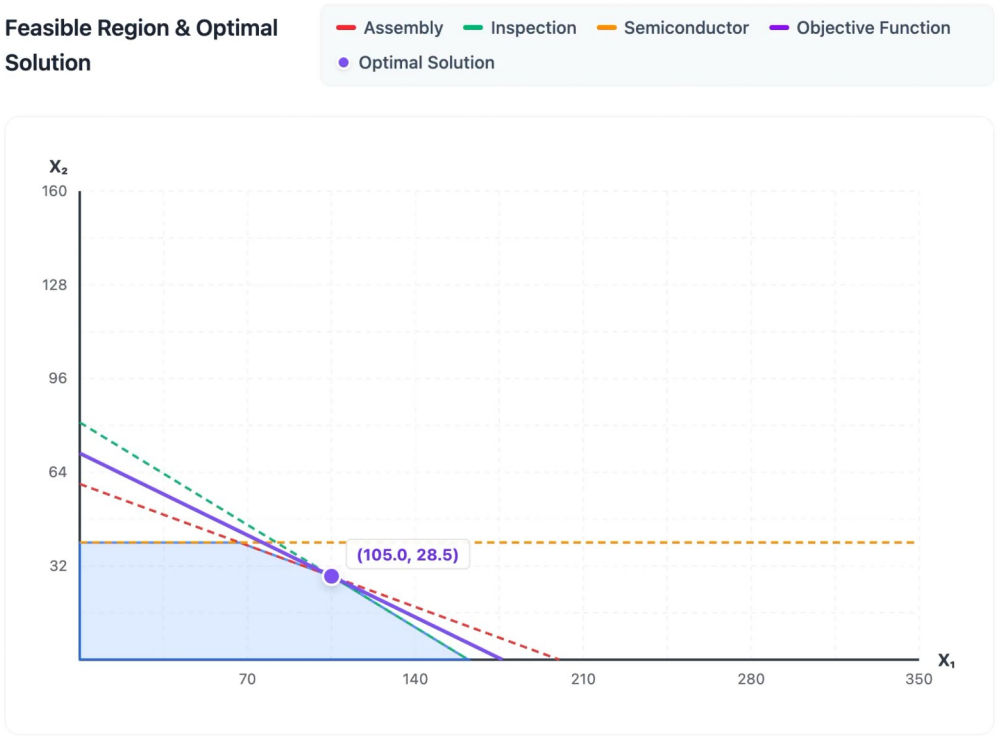
Inspection Dept. Available Hours (b_2) 81 hrs

$0.5X_1 + X_2 \leq 81$ — Shadow Price: $25.00

Semiconductor Chip Supply (b_3) 40 chips

$X_2 \leq 40$ — Shadow Price: $0.00

Feasible Region & Optimal Solution



Post-Optimal Analysis Using Sensitivity Analysis

Deciding whether to change the production plan (objective coefficient change)

- **Q: For what profit per standard lamp should we keep the current optimal daily plan (105 standard lamps, 28.5 high-performance lamps)?**

  (Hint) For 15 <= C_1 <= 25, the optimal solution (X_1=105, X_2=28.5) does not change

Post-Optimal Analysis Using Sensitivity Analysis

Deciding whether to change the production plan (objective coefficient change)

- **Q: For what profit per standard lamp should we keep the current optimal daily plan (105 standard lamps, 28.5 high-performance lamps)?**

  (Hint) For 15 <= C_1 <= 25, the optimal solution (X_1=105, X_2=28.5) does not change

- A: If the profit per standard lamp is between ₩150,000 and ₩250,000, the current plan (105 standard lamps, 28.5 high-performance lamps) can be kept. If it falls below ₩150,000 or rises above ₩250,000, the production plan must be revised

Post-Optimal Analysis Using Sensitivity Analysis

Deciding whether to add available hours (RHS change)

- Adding 1 hour of daily labor in the assembly process costs ₩50,000.
- Based on the current sensitivity analysis, should we add labor to the assembly process?
- If so, how much labor should we add?

(Hint) For $194.5 \le \mathbf{b_1} \le 258.4$, **the assembly shadow price stays at ₩62,500/hour**

Post-Optimal Analysis Using Sensitivity Analysis

Deciding whether to add available hours (RHS change)

- Adding 1 hour of daily labor in the assembly process costs ₩50,000.
- Based on the current sensitivity analysis, should we add labor to the assembly process?
- If so, how much labor should we add?

  (Hint) For $194.5 \le b_1 \le 258.4$, **the assembly shadow price stays at ₩62,500/hour**

- Net profit per hour: ₩62,500 – ₩50,000 = ₩12,500
- This shadow price is valid only for 194.5 ≤ b1 ≤ 258.4, so we can add at most 18.4 hours (258.4-240=18.4)
- Total additional profit = ₩12,500/hour × 18.4 hours = ₩230,000

Summary

Post-optimal analysis: what-if analysis performed after optimization

Shadow price	The increase in the objective function when a resource's supply capacity increases by 1
Reduced cost	The change in the objective function value when a variable that equals 0 in the optimal solution is increased by 1 unit
Sensitivity analysis	Analyzes how the optimal solution and optimal objective value change when model coefficients change Allowable ranges of objective coefficients and of the RHS